KOUSHIK HOWLADER

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PROFESSIONAL SUMMARY

Data scientist and computer science PhD student with experience in large-scale data analysis, machine learning, and data visualization. Skilled in Python, SQL, and analytical tool development for extracting insights from complex datasets. Experienced in building data processing pipelines, validating datasets, and developing visualization tools to support data-driven decision making. Interested in applying data analytics to real-world systems including industrial monitoring and operational analytics.

EDUCATION

- **PhD**, Computer Science, Iowa State University — *2023 - Present*
- **MS**, Computer Science, North Dakota State University — *2023*
- **MS**, Information Technology, Jahangirnagar University — *2015*
- **BS**, Information Technology, Jahangirnagar University — *2013*

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, R
- **Data Analysis:** Pandas, NumPy, Scikit-learn, Data Processing Pipelines
- **Visualization:** Matplotlib, Seaborn, Plotly, Power BI (familiar)
- **Tools:** Git, Linux, HPC Systems, Jupyter Notebook
- **Machine Learning:** Deep Learning, Transfer Learning, Predictive Modeling
- **Data Engineering:** Data validation, dataset monitoring, automated data processing

RESEARCH AND DATA SCIENCE EXPERIENCE

Iowa State University — **Iowa, USA**
Graduate Research Assistant – AI in Medicine — *2023 - Present*

- Designed and implemented machine learning pipelines to analyze large-scale biomedical datasets.
- Built data processing workflows for genomic datasets including mutation and copy number variation data.
- Developed analytical tools for ranking biological pathways and identifying patterns in large datasets.
- Created visualizations to interpret model outputs and communicate insights from complex data.
- Performed extensive dataset validation and preprocessing to ensure data integrity.

North Dakota State University — **North Dakota, USA**
Graduate Research Assistant – Machine Learning and Data Analysis — *2021 - 2023*

- Analyzed large-scale histopathology image datasets using machine learning and deep learning models.
- Implemented data preprocessing pipelines and automated data validation procedures.
- Conducted statistical analysis and performance evaluation of predictive models.
- Developed visualization tools to interpret model performance and dataset characteristics.

Center for Computationally Assisted Science and Technology — **North Dakota, USA**
Data Science Intern — *May 2023 - Aug 2023*

- Assisted interdisciplinary researchers in analyzing scientific datasets using HPC infrastructure.
- Optimized data processing workflows for large datasets on high-performance computing clusters.
- Supported troubleshooting of data processing pipelines and computational tools.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant** – Computer Science, Iowa State University — *2023 - Present*
- **Instructor and Teaching Assistant** – Computer Science, North Dakota State University — *2021 - 2023*
- **Assistant Professor** – Noakhali Science and Technology University — *2015 - 2020*

SELECTED PUBLICATIONS

- Transfer Learning Analysis on Cancer Histopathology Images. **BIBM 2022**
- Extracting Clinically Relevant Phrases from Patient Notes using BERT. **BIM 2023**
- Machine Learning Models for Detecting Type 2 Diabetes. **Health Information Science and Systems**

FELLOWSHIPS

- NSF Data Driven Discovery in Dynamics (D4) NRT Fellowship, Iowa State University — 2024–Present

PROFESSIONAL MEMBERSHIPS

- IEEE Member
- ACM Member